Rongali Mohit Naidu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.23
Sri Chaitanya Jr College Vijayawada	Class XII (BIEAP)	2023	96.10%
Sri Chaitanya High School Dwarakanagar	Class X (SSC)	2021	99.83%

POSITIONS OF RESPONSIBILITY

UG Mentor at Sunshine Counselling Cell 2024-2025

SKILLS

Programming Skills: C/C++ ,Python.

Web Technologies : HTML, CSS,Node.js

Database Management: MYSQL,Redis, AWS S3

Tools: Git, Latex, Solid Edge

RELEVANT COURSES

CSE Core Courses: Intro to Programming, Software Development Fundamentals, Theory of Computation, Operating Systems,DBMS, Computer Architecture, Data Structures, Algorithms.

Mathematics and EE Courses: Discrete Mathematics, Calculus, Digital Circuits, Probability and Random Variables

EXTRACURRICULAR

Cricket Team Member, IITH Hostel League :

Represented hostel in inter-hostel cricket tournaments; contributed to team strategy and coordination. Volunteer,

NSS (National Service Scheme), IITH :

Participated in organizing blood donation camps, plantation drives, and literacy awareness initiatives.

PROJECTS

Build an AI gateway with cost-quality routing, safe caching and scalable model serving

- Adaptive LLM Routing**:** Built a policy layer to dynamically route requests across heterogeneous backends based on prompt/output length, cost, and latency targets.
- Two-Stage Request Routing**:** Decoupled model classification from backend replica selection to balance quality targets with real-time latency.
- Benchmarking & Simulation**:** Evaluated adaptive routing against Round-Robin and Least-Loaded baselines under bursty and fault-injected workloads

Document Management Repository (DMR)

- AI Document Ingestion & Routing: Integrated Groq API and pdf-parse to extract metadata and dynamically classify and route incoming PDFs into keyword-targeted vaults.
- High-Throughput Cloud Pipeline: Built a zero-memory streaming pipeline utilizing zlib and AWS S3 SDK to compress text formats on-the-fly and cached pre-signed URLs via Redis to minimize latency and database load.
- Access Control & Multi-Tenancy: Implemented a granular Role-Based Access Control (RBAC) system with JWT and

hashed developer API keys across public, private, and multi-tenant organizational workspaces.

Multi-Client Quiz System with LLM Integration:

- Multi Threaded TCP Architecture: Built a C/C++ quiz application handling up to 10 concurrent clients using a custom application protocol and KEEP-ALIVE heartbeat detection.
- LLM Integration & Caching: Integrated an LLM API backend for real-time question generation with an in-memory genre cache to reduce latency.

Synchronization Tools Analysis (Course Work / Prof. Sathya Peri)

- Evaluated performance of Test-and-Set (TAS), Compare-and-Swap (CAS), Bounded Waiting CAS, and Atomic-Increment algorithms based on average and maximum thread completion times for critical sections for mutual exclusion in parallel computing in C++.
- Analyzed their performance based on average and maximum time to enter the critical section.

Multi-threaded Web Crawler

- Engineered a multi-threaded web crawler using a producer-consumer architecture, shared URL queues, and thread synchronization to concurrently crawl and process web pages.
- Implemented fault-tolerant checkpoint recovery with MySQL by persisting visited/pending URLs and crawl statistics, allowing interrupted crawls to resume without restarting.
- Built an LLM-powered Text-to-SQL assistant with LangChain that converts natural-language queries into SQL, retrieves database results, and generates formatted responses through a React interface.